

Part I

General Review

This page intentionally left blank

Chapter 1

Introduction

Any news concerning central bank policy decisions always garners public attention. Statements from the governor of the central bank in the United States, the Chair of the Federal Reserve System (the Fed), or indeed other members of the Federal Open Market Committee (FOMC), about the Federal Funds Rate (FFR), for instance, are always eagerly awaited by markets around the world.¹ In fact, indications of the FFR direction, which are typically linked to statements from the Chair of the Fed or other FOMC members about US inflation and the economy, usually become a source of market speculation. Such conditions have become unequivocal since the Fed announced in May 2013 its plan to normalize the ultra-loose monetary policy stance adopted after the global financial crisis (GFC) of 2008/2009. This stance was to support the US recovery, which subsequently became known as the Fed's Taper Tantrum. Actual decisions, or even just indications of the policy rate direction, directly influence money market interest rates, dollar exchange rates on the foreign exchange market, and stock prices on Wall Street. Such developments lead to reactions of interest rate, exchange rate, and stock prices in a number of countries, including Indonesia.

Similarly, in Indonesia, statements released by Bank Indonesia (BI) are constantly in the news across various mass media outlets. The decisions of the BI Board of Governors regarding the BI Rate or even just indications about the possible future direction of the policy rate are increasingly becoming a reference for the markets and banks in terms of the financial transactions undertaken.² In practice, non-resident investors, specifically those that invest in financial

¹See, for example, the "Transcript of Chair Yellen's Press Conference," December 16, 2015, Board of Governors of the Federal Reserve System (2015), <http://www.federalreserve.gov>. At the press conference, Chair Janet Yellen relayed the FOMC decision to hike the FFR by 25 bps, from 0.25% to 0.50%. This follows seven years of a near 0% rate in order to support the US recovery in the wake of the worst GFC in the United States since the Great Depression in the 1930s.

²Pursuant to the BI announcement, dated April 15, 2016, effective from August 19, 2016 the BI Rate, which is equivalent to the 12-month money market rate, will be replaced by the BI 7-day (Reverse) Repo Rate, representing the seven-day money market rate.

instruments such as government bonds (SUN), stocks, BI Certificates (SBI), and corporate bonds in Indonesia, base their investment decisions on statements regarding the BI Rate. In addition to policy rate decisions, the BI policy in relation to macroprudential regulation and supervision, including down payments on automotive loans as well as the maximum loan-to-value (LTV) ratio on property and housing loans, also attracts public attention. So too does payment system policy, such as money supply, clearing, card-based payment instruments, and electronic money. Decisions relating to the policy rate, macroprudential policy and the payment system are based on rigorous assessments of global and domestic macroeconomic and financial system developments and projections that are regularly delivered by BI.³

1.1. The Central Bank and Economy

The brief overview in the preceding section demonstrates the importance of the central bank's role in the economy. Since inception, the central bank has been mandated to maintain domestic currency stability (inflation and exchange rates) as well as financial system stability. Maintaining currency and financial system stability is the primary contribution of the central bank toward supporting sustainable economic growth. Therefore, the functions of monetary and payment system policy, as well as the regulation and supervision of financial institutions, fall under the auspices of the central bank. Initially, such functions were comparatively simple but have become increasingly complex over time in line with global and domestic economic and financial development. This can be gauged from the changing role of the monetary sector, from the commercial paper predominance (rediscounted by the Bank of England in the seventeenth century) to monetary operations targeting interest rates. The function of the payment system, on the other hand, has also progressed from merely printing and circulating banknotes and coins to the regulation and supervision of payment instruments, mechanisms, and infrastructure. Likewise, in terms

³See, for instance, the BI Press Release, dated December 17, 2015, that announced the BI Rate would be held at 7.50% in anticipation of pressures from global financial markets, including the impact of the FFR hike. This resulted despite stable macroeconomic conditions at home. Indonesia, for instance, recorded low inflation and a sustainable current account deficit, which provided adequate space to ease the monetary policy stance. The BI Rate was subsequently lowered at the BI Board of Governors' Meeting in January, February, and March 2016 by a total of 75 bps to 6.75%. At the BI Board of Governors' Meeting in June 2016, in addition to reducing the BI Rate by 25 bps to 6.50%, BI also introduced macroprudential policy easing by raising the LTV ratio to boost lending and economic growth. In addition to news in the mass media, the decisions of the BI Board of Governors are published as press releases and posted to the official website of BI at <http://www.bi.go.id>. The BI also regularly publishes a quarterly Monetary Policy Report and monthly Monetary Policy Review that contains assessments of economic, financial, and monetary developments and projections that underlie BI Rate policymaking.

of financial system stability, the function of the central bank has progressed from been a lender of last resort (LOLR) to macroprudential regulation and supervision.

The monetary policy instituted by the central bank to influence money supply and interest rates is one determinant of macroeconomic stability; in particular, inflation and exchange rate stability. In the financial sector, the interest rate policy and exchange rate stabilization policy of the central bank directly influence bank funds and credit, as well as stock and bond prices on the capital market. Subsequently, through the influence of such monetary and financial developments on consumption, investment, exports, and imports, monetary policy also influences inflation, economic growth and, therefore, the creation of employment opportunities in addition to the balance of payments. In other words, the monetary policy stance adopted by the central bank has a major influence over public prosperity and welfare.

Similarly, microprudential and macroprudential regulation and supervision are imperative in terms of maintaining financial system stability. Microprudential regulation entails microregulation and supervision of financial institutions and focuses on the soundness and performance of each individual financial institution. Meanwhile, macroprudential regulation involves macroregulation and supervision of financial institutions and focuses on systemic risk in order to achieve financial system stability. In many countries, macroprudential and microprudential regulation and supervision fall under the auspices of the central bank. In contrast, several countries, including Indonesia, Australia, and South Korea, transferred the microprudential function to a financial services authority, while the central bank maintained control over macroprudential aspects. In its implementation, central banks direct macroprudential policy toward dampening accelerators in the financial cycle, while preventing and detecting a build-up of systemic risk that leads to financial system instability. Regulations concerning the LTV for housing loans as well as down-payments on automotive loans in Indonesia are concrete examples of macroprudential policy to combat excessive credit growth in both sectors that could disrupt financial system stability.

In terms of the payment system, currency is printed and circulated in line with the economic requirement and, therefore, supports monetary policy to achieve price stability and macroprudential policy to maintain financial system stability. In addition, payment system policy includes the reliable, efficient, and secure transfer of funds, clearing and financial transaction settlement, retail and wholesale, in the economy. Various payment instruments have also been developed in line with the advancement of financial product innovation and information technology development, such as automated teller machines (ATM), debit and credit cards, mobile and internet banking as well as electronic money. Even today, in the era of technological progress, the development of financial technology, or FinTech, has fundamentally changed the business models of various financial services offered by financial institutions to the public. Therefore, payment system development, monetary and financial system stability will affect each other, with all three considered crucial for economic advancement.

Due to the significant influence policy has on the economy and public welfare, it is understandable that the public, business, and financial communities, as well as government and parliament take an avid interest in the central bank. That interest extends beyond policy aspects to the institutional arrangements of the central bank in line with the ongoing change in the global economic and political landscape, with more and more countries applying a market-based economy and democratically elected governments. The institutional arrangements of the central bank are reinforced through the principles of good governance by strengthening the legal framework and in terms of policy implementation. This can be seen by the modernization of prevailing laws that provide a clear mandate for the central bank as well as independence in the execution of its duties. Furthermore, greater accountability and transparency from the central bank is now in increasing demand in terms of policy implementation. The various institutional arrangements of the central bank have been an ongoing concern for the past two decades and became increasingly important in the wake of the GFC. These changes reflect growing public awareness in various countries of the need to strengthen the role and standing of the central bank in order to support achievement of the economic policy targets.

1.2. Central Bank, Academic Thinking, and Political Economy

From a central bank standpoint, the challenges faced in terms of carrying out the mandate are onerous. In fact, the challenges have become even more complex since the onset of the GFC. In the implementation of monetary policy, for instance, maintaining low and stable inflation has become increasingly important to support economic growth and ameliorate public welfare. The volatility of non-resident capital flows and exchange rates in emerging market economy (EME), including Indonesia, have also increased since the GFC with monetary policy divergence stemming from ultra-loose monetary policy in advanced countries and ubiquitous uncertainty blighting global financial markets. The GFC also showed that price and exchange rate stability alone are insufficient to maintain financial system and macroeconomic stability to support sustainable economic growth. Increasingly rapid and complex development in terms of product innovation and financial operations on the one hand has facilitated economic financing but, on the other hand, has also amplified the risk of financial system instability and affected the monetary policy transmission mechanism in the economy. Likewise, payment instrument development has necessitated increased regulation and supervision by the central bank in order to maintain a reliable, efficient, and secure payment system. Institutionally, the strengthening of consistency, independence, and accountability as well as transparency and communication have become progressively more important, not only in terms of governance but also to support policy effectiveness and credibility and, therefore, the reputation of the central bank.

Throughout their evolution, economic conditions, political environment, and academic thinking have influenced how a central bank implements its mandate.

At its inception in the seventeenth century, for instance, the role of the central bank in the economy emphasized creating and circulating currency, purchasing government debt, and functioning as LoLR for financial system stability. Application of the gold standard strengthened central bank credibility in terms of achieving price, exchange rate, and financial system stability because the central bank was charged with maintaining currency convertibility in line with the gold reserves held. Nonetheless, government control over the central bank to finance the post-World War recovery triggered soaring inflation, leading to an economic crisis. Furthermore, central bank credibility was also lost.

Developments over the two decades prior to the GFC saw central banks focus shifted on price stability. This was in response to soaring inflation, while providing the central bank independence from government control. On the other hand, however, the focus on price stability was also based on the growing acceptance of New Neoclassical and New Keynesian synthesis, namely that monetary policy only influences inflation in the long term, despite a short-term trade-off between inflation and economic growth in line with the findings of the Phillips Curve. The economy is assumed to always be in equilibrium and the primary causes of imbalances are price and wage rigidities. Similarly, there is no friction in the financial system, therefore currency and credit are perfect substitutes that are affected by interest rates. Furthermore, non-resident capital flows freely and, therefore, a fully flexible exchange rate system is the optimal choice. Consequently, central bank policy merely needs to focus on stipulating the short-term interest rate in order to achieve price stability, while economic and financial equilibrium will generate economic growth and maintain macroeconomic and financial system stability. Central bank policy governance is strengthened by independence and is in line with rational expectations theory, policy consistency with the rules and the importance of transparency to form and anchor expectations. Such theories and ideas compelled central banks in many countries to target price stability, thereby adopting the Inflation Targeting Framework (ITF).

The GFC, however, turned central bank practices and theories on their head; not because ITF policy had failed but, in contrast, because ITF had successfully lowered inflation in many countries, coupled with low interest rates that had stimulated rapid economic growth. The problem was that long-term stability and economic boom had led to excessive credit growth, asset price bubbles (stocks and housing) as well as high leverage. Financial accelerators caused the financial cycle to amplify the economic cycle. Furthermore, economic stability led to financial system instability that ultimately culminated in the worst crisis (namely, the GFC) since the Great Depression of the 1930s. Evidently, price stability alone is insufficient to ensure macroeconomic stability if financial system stability is not also maintained; “there is no macrostability without financial stability.” In reality, financial friction is inevitable due to asymmetric information, financial product innovation, price setting, and valuation as well as risk-taking behavior, hence the financial system is constantly in a state of flux that produces financial accelerators and turns fragility to systemic risk.

Consequently, the GFC taught an invaluable lesson that the central bank should return to its original mandate, namely to achieve and maintain the stability of

domestic currency values (inflation and exchange rate), while supporting financial system stability. Central bank credibility and the framework that had been established through ITF became the foundation to achieve that mandate. Nevertheless, that was still not enough. Macrofinancial linkages demanded a macroprudential policy response from the central bank to mitigate procyclicality between the financial sector and economic activities that trigger economic and financial crises, such as the GFC. The policies of foreign exchange market intervention and non-resident capital flow management were also required to stabilize the exchange rate, while remaining in a flexible regime. Such developments encouraged many central banks to apply macroprudential policy and manage foreign capital flows in order to strengthen the effectiveness of monetary policy. In brief, the monetary and macroprudential policy mix became the new central banking paradigm in various countries after the GFC, including BI in 2010. This innovation in central bank policy practices also demands the development of underlying theories and empirical studies.

The previous description illustrates that the theory underlying central bank policymaking has developed with rapidity. Likewise, the practice of policy formulation and implementation at various central banks has experienced a paradigm shift that requires the development of underlying theory. In other words, the advancements have had an advantageous reciprocal influence between the development of financial and monetary economic theory in the academic world and the various schools of thought that underlie practical central bank policymaking. Academia, in the pursuit of clarifying or offering solutions to prevent problems, has contributed greatly to conceptual thinking and theories concerning various phenomena and economic behaviors. The theoretical constructs have become advanced and deep on a philosophical conceptual level and using quantitative methods along with empirical studies, although in this case relying on certain assumptions that simplify complex economic behaviors.

On the other hand, the central bank has also made various breakthroughs in terms of conceptual ideas and innovative policymaking as a solution to complex financial-economic problems. Oftentimes, policy innovations are also facilitated by close interaction between the central banking community at various global and regional forums and meetings. Theoretical development in academia is clearly a solid foundation for central bank policymaking. Nonetheless, the complexity of existing problems often necessitates an innovative policy response from the central bank, which also encourages the academic world to conduct theoretical studies and seek underlying empirical evidence.

Rapid development in terms of policy concepts and institutional arrangements at the central bank, from the perspectives of both academia and the innovative policy response of the central bank, demands stringent reviews, studies, and documentation. The various concepts underlying the policy response of the central bank are a vital reference for the central bank and other policymakers to draw lessons on the general and best practices available to prevent the problems faced as well as for academia to seek clarification and develop the theories further. On the other hand, the relentless development of theories and ideas in academia are a helpful reference for the central bank in terms of policymaking in response to emerging problems.

For academia, the synthesis (or indeed contradictions) that may appear between central bank policy theories and practices represent an interesting study that must be taught to students in order to enhance academic knowledge or, indeed, for further development. This is important to bridge the gap between theoretical ideas in the academic world and the practical policymaking process at the central bank. Moreover, academia, especially in Indonesia, has limited access to advanced books that discuss central banking policy theories and practices as a reference for postgraduate students, teachers, and researchers. In general, existing programs and subjects tend to focus on standard theories, which lack comprehensive and informed discussions on central bank policy practices and the underlying theories. Similarly, reviews of institutional aspects are also absent from academia.

1.3. Objectives of the Book

This book aims to analyze various policy theories and practices adopted by central banks as well as the institutional arrangements underlying the principles of good governance in policymaking. In other words, this book explores the synthesis between theories, practices, and institutional arrangements that underlie central bank policy. The discussion focuses on philosophical conceptual theories and the policies themselves, supported by relevant quantitative analyses as required along with various issues and the latest problems as they emerge. This book aims to bridge central bank policy theories and practices at the intermediate and advanced levels, which, according to the authors, is currently lacking and, therefore, will provide an invaluable contribution to the world of knowledge.

The goal is to produce a reference book on central bank policy theories and practices in order to advance the understanding and insight of policymakers and practitioners, as a study resource for researchers and academic teaching materials for teachers and students. To that end, the material discussed and contained in this book represents a combination of three perspectives. *Firstly*, a philosophical review and quantitative analysis of the theories underlying central bank policymaking. *Secondly*, the policy practices employed by central banks in various jurisdictions, including Indonesia. *Thirdly*, a study on the influence and implications of central bank policies and regimes to economic, monetary, and financial development in the countries concerned. The various empirical studies presented substantively explain and enrich the three dynamic perspectives.

Admittedly, it is not a simple task to author and present such a reference book that bridges policy theories and practices at the central bank. Several challenges forced the authors to review and consider the most appropriate material that represents the overarching purpose. Onerous challenges were encountered not only due to the rapidity of theoretical and practical development, which have already been discussed in various other books, journals and papers, but also because of the panoply of central bank policy practices that do not yet appear in print as a reference for the writing of this book. In this case, the authors benefitted from extensive experience and direct involvement in the policymaking process at BI and various international forums, as well as academic activities linked to various

postgraduate programs at several universities in Indonesia. The authors seek to review various theories and constructs under development that are considered relevant and most clearly color the policy practices at the central bank, without rehashing existing discussions in more detail that can be found in the existing literature on monetary economic theory as taught in academia.

During the writing process, one of the largest challenges was sorting the relevant theories for policymaking as practiced. More specifically, versatility when exploring the plethora of best policy practices employed by various central banks, while simultaneously raising the variety of debates and critiques under development, was imperative. This challenge stemmed not only from the comparatively new policy practices, for example, the Flexible Inflation Targeting (FIT) Framework, macroprudential policy or the institutional aspects of diverse central banks but also the review of empirical evidence concerning how policy and institutional arrangements contribute propitiously to central bank and economic performance.

1.4. Systematics of the Writing

The systematics of this book are based on the relevance of policy theories and practices at central banks that the authors consider important for policymakers, practitioners, researchers, and as academic teaching materials. In addition to the experience gained from teaching postgraduate programs, the authors' involvement in the policymaking process at BI, coupled with the rare opportunities to attend seminars, conferences, and a broad range of meetings with other central bankers in numerous jurisdictions are used as a key reference in the writing of this book. Through a process of contemplation, the authors ultimately decided to divide the book into five inseparable parts as the systematics of the writing.

Part I explores the evolution, reforms, and latest issues concerning central bank policy, while Parts II–V discuss in depth the core of the material about central bank policy theories and practices. Those four parts include a theoretical review of the influence of central bank policy on the economy in Part II, central bank policymaking theories and practices in Part III, as well as the institutional theories and practices that underlie the principles of good governance in terms of central bank policy in Part IV. Part V of the book presents the latest perspectives and new paradigm for central banks in the wake of the GFC, namely the central bank policy mix consisting of monetary policy (including foreign capital flow management), macroprudential policy, and payment system policy, to achieve the dual mandate of price (and exchange rate) stability as well as financial system stability.

Referring to the evolution and reformation of central banks since their inception in the seventeenth century up until the aftermath of the GFC, a number of important lessons can be given. *Firstly*, the central bank plays a significant role in supporting sustainable economic growth through price (and exchange rate) stability, while creating and maintaining financial system stability. This dual mandate has existed since the formation of central banks and was tangibly reinforced by the GFC. *Secondly*, in pursuance of the dual mandate, a central bank requires

jurisdiction over monetary policy (including foreign capital flow management), the payment system as well as macroprudential regulation and supervision. Experience in the ITF era has shown that a focus on monetary policy by central banks contributed to financial instability that culminated in the GFC. *Thirdly*, central banks must be given autonomy to implement their authority in pursuit of the goals or mandate. Experience from when the government exercised control over the central bank also showed the adverse impact on macroeconomic stability and ultimately resulted in economic crisis. *Fourthly*, institutional arrangements need to be strengthened to ensure that central bank independence is accompanied by public accountability and transparency.

Part II of this book, consisting of Chapters 3–5, reviews those theories considered relevant and lay the groundwork for the central bank policy practices. Chapter 3 discusses monetary economic theory in a closed economy, stressing the role of money and credit in the economy. Concerning the role of money, the authors put forward a debate on the neutrality versus non-neutrality of money from the perspective of Classical Theory and Keynesian Theory, the models that were subsequently developed as well as the empirical evidence for the influence of money in the economy. The theoretical discussions lead to a synthesis that money supply exerts more influence on prices in the long term, while acknowledging the trade-off between inflation and growth in the near term. This chapter closes with the new paradigm of monetary economic theory developed by J. E. Stiglitz, which emphasizes monetary analysis of credit quantity, not money! Therefore, monetary policy instituted through money supply or interest rates must be strengthened by banking regulations that directly affect credit, particularly through macroprudential policy. A monetary and macroprudential policy mix is required to support price and financial system stability.

Chapter 4 takes the reader into monetary theory space for an open economy by exploring several theories that tackle the role of the exchange rate in the economy. The discussion focuses on studying models that determine exchange rates, which are discussed ad nauseam in the literature, from the Mundel–Flemming model, to the flexible price monetary model, Dornbusch’s overshooting model and the money substitution model. Various models and empirical evidence concerning the influence of the exchange rate on the economy, especially in terms of predicting exchange rate behavior and the effect of exchange rates on international trade, economic growth and inflation, are also explored. Specifically, a discussion is presented on the monetary policy trilemma in an open economy, namely between monetary policy independence for domestic purposes (price stability and economic growth), foreign capital flow mobility and exchange rate flexibility. The chapter closes by identifying some implications concerning the influence of exchange rates on monetary policymaking by the central bank, namely the requirement for an interest rate policy, foreign exchange market intervention, and capital flow management to overcome the aforementioned policy trilemma.

Chapter 5 discusses the monetary policy transmission mechanism, in other words how monetary policy instituted by the central bank affects the financial system and economy. This represents a critical aspect underlying monetary

policymaking at the central bank in order to achieve the desired target but which, according to the authors, receives insufficient theoretical and empirical attention. The discussion also explores the two schools of thought regarding how monetary policy is transmitted to the economy, namely the money view by which policy is transmitted through the interest rate channel, exchange rate channel, asset prices, and expectations, as well as the credit view by which policy is transmitted through the bank credit and capital channel, the balance sheet and collateral constraints channel as well as the risk-taking behavior channel. In addition to theoretical discussions, various empirical studies are also explored concerning the workings of the monetary transmission mechanism in several countries, including advanced countries, EMEs, and Indonesia itself. It will be argued that the effect of monetary policy on price stability and economic growth is indirect and experiences a lag. In general, it will be argued that the channels of the money view continue to function as the principal transmission mechanism. Nonetheless, the GFC demonstrated the growing importance of the credit view channels due to financial frictions that stem from asymmetric information, financial product innovation and risk-taking behavior. Therefore, central bank monetary policy based on interest and exchange rates must be backed up by policy to manage credit through macro-prudential policy.

Part III of the book, containing Chapters 6–9, details the central bank policy framework applied during the two decades prior to the GFC. Like the explanation presented in Chapter 2, during this period, central banks tended to focus more on monetary policy to achieve price stability in support of economic growth. In many ways, the policy framework was a success and remains the backbone of central bank policy in various countries to this day. That framework underlies the theoretical and empirical discussions contained in Part II. Chapters 6 and 7 focus on the strategic and operational frameworks of monetary policy. Thereafter, Chapters 8 and 9 discuss the monetary policy framework with price stability as the overarching goal, which is practiced widely around the world, including in Indonesia, under the popular ITF.

The discussion on the strategic monetary policy framework in Chapter 6 focuses on the strategic issues regularly debated to this day, namely the “duality” of accomplishing the ultimate monetary policy target: output stability (economic growth) or price stability (inflation)? Opening with the debate between economic growth and price stability as the final target of monetary policy, several policy regimes practiced by central banks as well as on a theoretical level are explored, such as exchange rate targeting, monetary targeting, inflation targeting, and nominal gross domestic product (GDP) targeting. Empirical evidence showing what such regimes are based on and the implications to economic performance in various countries is presented considering the historical perspective and evolution of policy orientation over time. In addition, several issues relating the Phillips Curve are also interpreted from the perspective of short- and long-term linkages between inflation and economic growth as well as the socio-economic costs (sacrifice ratio) that arise in the price and output stabilization process.

Chapter 7 elaborates the monetary policy operational framework in terms of two other “duality,” namely: (1) Which are the preferable operational targets or

policy instruments used to achieve the ultimate target of existing policy? Which are the quantity-based instruments or price-based instruments? (2) Which is the preferable policy response should be formulated; a rule-based response or discretion? The discussion commences with the selection of these monetary instruments to address various economic shocks; specifically, a distinction is made between demand shocks or supply shocks. With the operational target selected, the follow-up issue addressed is the optimal policy response to achieve the final target set. This chapter also presents a hypothesis for the existence of the state-contingent rule, which can accommodate rules and discretion, as an optimal policymaking response, particularly in the case of EMEs.

As the most popular monetary policy framework adopted by central banks, the ITF discussion delivered in Chapters 8 and 9 focuses on its application at the central bank, including a case study of Indonesia. Institutional and operational regulations as well as their impact on central bank and economic performance are also explored rationally. Several ITF regimes practiced at central banks are also discussed and compared with other monetary policy regimes. The discussion in this chapter also addresses a number of misconceptions that persist at the central bank itself or in academia, such as how economic growth, exchange rates, and other aspects are accounted for under ITF. Part V of the chapter presents an assessment of ITF application in several developing countries and provides justification for the adoption of a FIT framework as the ideal format for a small open economy.

Part IV of the book, incorporating Chapters 10–12, focuses on the salient aspects of implementing good governance in terms of the institutional arrangements and central bank policy. In addition to the monetary policy practices of central banks explored in Part III, the institutional aspects discussed in Part IV remain elusive and rarely taught academically. This is not only because such aspects include contemporary issues of central bank policy but also because it is important to discuss such issues in the context of the need to strengthen public institutions in line with the spread of democracy as well as policy transparency and accountability.

Chapters 10 and 11 explore the institutional aspects salient to monetary policy, namely building policy credibility and striking the optimal balance between independence and accountability. Based on time inconsistency theory, in accordance with the models developed by Kydland–Prescott and Barro–Gordon, the discussion in Chapter 10 stresses the need for the central bank to follow a certain rule when formulating policy, for example, the Taylor rule, rather than discretion. The discussion continues with a number of approaches found in empirical studies to measure credibility as well as the contribution to central bank monetary policy performance with respect to supporting economic performance, including an assessment of monetary policy credibility in Indonesia. Chapter 11, meanwhile, presents the delegation of authority theory using the Rogoff model that emphasizes the importance of delegating authority by parliament to a conservative central bank that is inflation averse in order to stipulate an achievable inflation target. The theory is based on central bank independence since the 1980s in terms of policymaking to achieve the mandate, namely price stability. Central

bank independence also requires strong accountability to ensure credibility is achieved.

Chapter 12 focuses on the growing importance of transparency and communication as an integral part of institutional strengthening and good governance at the central bank. This is not only increasingly demanded by the public as democracy flourishes but also because transparency and communication have become crucial instruments that support central bank policy credibility and effectiveness. This chapter shows that the phenomenon of increased policy transparency will persist moving forward and become a main menu item for central banks in the implementation of monetary policy. Therefore, transparency must be evaluated and strengthened coupled with a clear monetary policy communication strategy referring to international best practices, the monetary policy regime adopted and financial market development in the respective jurisdiction.

Part V of the book, namely Chapters 13–15, presents a new paradigm that has evolved as central bank policy theories and practices since the GFC. Two important observations have emerged since the GFC. *Firstly*, the importance of the central bank returning to its original mandate at inception, namely price (and exchange rate) stability, while creating and maintaining financial system. *Secondly*, the need for monetary policy through interest rates to be reinforced by exchange rate stabilization policy and foreign capital flow management in an optimal policy mix along with payment system policy and macroprudential policy. The GFC demonstrated the need for the central bank to prevent crises through macroprudential regulation and supervision in order to maintain financial system stability from the perspective of macrofinancial linkages and systemic risk. In addition, the GFC also heightened the volatility of foreign capital flows from global financial markets to EMEs, thus necessitating more than just interest rate policy, namely exchange rate stabilization policy and foreign capital flow management to address the ever-present central bank policy trilemma. An optimal policy mix between monetary policy (including foreign capital flow management), payment system policy and macroprudential policy represents an innovative new central banking paradigm to achieve the dual mandate of price stability and financial system stability.

Chapter 13 discusses central bank policy complexity and the response to heightened exchange rate and foreign capital flow volatility observed since the GFC, referring to the discussion presented in Chapter 4. The discussion begins by analyzing the push and pull factors of foreign capital flows to a country and the implications of volatility on monetary and financial system stability. The discussion then moves on to focus on the policy trilemma of achieving price stability, exchange rate stability, and free flows of foreign capital. The overarching message from the discussion in this chapter is that the policy response of the central bank, which is generally based on interest rates, must be reinforced by foreign exchange market intervention and foreign capital flow management in order to achieve price and exchange rate stability. The two-target approach (price and exchange rate stability), using two instruments (interest rates and foreign exchange market intervention), should strengthen central bank policy effectiveness and credibility.

Chapter 14 discusses the role of the central bank in terms of financial system stability through macroprudential policy. The discussion reiterates the importance of financial system stability as a costly lesson of the GFC, and how the central bank supports financial system stability using macroprudential policy. Theoretical and empirical studies underlying two important areas of macroprudential policy are explored. The first relates to the phenomenon of financial accelerators as a characteristic of the financial sector, which tends toward procyclicality, as well as excessive credit growth and asset bubbles. The second relates to interconnectiveness and networking in the financial sector, such as that found on the interbank market and in the payment system, which could exacerbate and transform contagion risk into systemic risk that endangers financial system stability. The discussion also extends to the implementation of macroprudential policy frameworks in various countries, including advanced countries, EMEs and Indonesia itself, in terms of the principles, targets, and instruments employed. The chapter closes with a discussion of the importance of coordination between macroprudential policy and microprudential policy as well as macroeconomic policy (monetary and fiscal) in strengthening financial system stability.

The new central bank policy paradigm is discussed in depth in Chapter 15. The discussion covers the conceptual dimensions of the central bank policy mix, including the integration of price stability and financial system stability, the mix of policy instruments used, as well as the transmission mechanism. Thereafter, the discussion focuses on two modeling approaches for the central bank policy mix. The first is the macroeconomic structural model known as IIF, which is an offshoot of FIT, covering the structure and analysis framework, as well as the central bank policy mix formulation model. The second is the Dynamic Stochastic General Equilibrium (DGSE) model, which expands the macroeconomic analysis with macrofinancial linkages based on financial frictions to integrate monetary and macroprudential policy when formulating the central bank policy mix. The chapter goes on to explore the policy mix instituted by BI as a new paradigm initiated in 2010, encompassing the targets and instruments, formulation of the interest rate, exchange rate, and macroprudential policy along with foreign capital flow management as well as strengthening institutional arrangements and coordination with the Government and other relevant authorities.

As mentioned previously, the series of discussions in the chapters of this book are fundamentally inseparable. In that context, after the explanation of central bank evolution and reform since inception in the seventeenth century until after the GFC in Chapter 2, the authors deepen the theories on the role of money and credit in the economy, foreign capital flow mobility, and exchange rates, as well as the monetary transmission mechanism in Chapters 3–5. The policy framework based on ITF discussed in Chapters 6–9 becomes the subsequent foundation of good governance in the implementation of monetary policy and the implementation of the central bank policy mix. Likewise, institutional reform through independence on the one hand and strengthening accountability, transparency, and communication on the other, which is discussed in Chapters 10–12, becomes a fundamental step in terms of enhancing policy credibility and central bank reputation. To close, the modern central bank policy mix applied in the wake of

the GFC is discussed in Chapters 13–15 theoretically and as practiced at various central banks. It is important to note that the policy mix represents a new central banking paradigm since the GFC, and BI should be recognized as a pioneer in terms of its implementation since 2010.

1.5. Utility of the Book

This book is considered a first in terms of comprehensively discussing the latest theories and practices of central bank policy. With the scope of material presented and systematics of writing employed, this book is believed to be a practical reference tool for policymakers at the central bank, practitioners, and academia. For the leaders, researchers, and staff at BI, as well as the financial-banking and business communities in general, the discussions contained in this book, especially in Parts III and IV, are fundamental to deepen the various concepts of central bank policy and theory. Similar benefits will also be enjoyed by economists and entrepreneurs in other sectors. Although the discussions are predominantly on a philosophical and conceptual level, accompanied by in-depth theory, the rationale, and writing of this book is presented in a manner that is readable and easily digestible.

For academia, this book will be particularly useful as a primary and essential reference for university students of monetary and financial economics at the intermediate and advanced levels of Master's degrees and Doctoral degrees. Meanwhile, for university students at the senior level of Bachelor's degrees, this book will also be very useful as a reference on economic modeling and empirical evidence regarding the central banking issues discussed. Furthermore, this book will also be suitable as a foundation for the development of a postgraduate central banking curriculum. Besides, the book will also be particularly relevant for teachers and students to deepen central bank policy and theories in the central banking curriculum or, indeed, for other applicable subjects, such as the monetary sector and financial institutions. As stressed previously, this book will furnish teachers and students alike with a comprehensive and deep understanding of relevant economic theories, policy practices, and institutional aspects supported at the central bank.

As a primary reference for financial and monetary economics students as well as postgraduate central banking programs, this book was designed to be taught in one full semester. The authors' experience of teaching such subjects at postgraduate programs underlies that belief. Moreover, the chapters of the book are also presented in an order such that it aids the learning process for students. Nonetheless, each respective teacher maintains the scope to explore certain aspects in more or less depth as preferred. Students, including those taking monetary and financial economics or similar subjects at university, as well as those exploring central bank policy to write a thesis or even to deepen insight into economic policy, will have no problem understanding the materials and concepts put forward in this book.

Over time, the materials presented in this book will be adjusted according to the prevailing requirements and developments in terms of policy practices at central banks as well as input from the readers in line with the requirements brewing among the public and in academia. To that end, inputs from readers are warmly welcomed for further refinement and advancement of this book. In closing, the authors sincerely expect this book to not only meet the needs of the readers in terms of a deeper understanding of the various salient aspects linked to central bank policy and theories but also to expand our scientific understanding and knowledge. Hopefully!

This page intentionally left blank

Chapter 2

Central Bank Evolution and Reform

2.1. Introduction

The GFC reconfirmed the vital economic role played by the central bank. Each time a crisis event transpires, the central bank is the most active organization in the stabilization efforts to resolve the crisis and recover the economy. During the GFC, for example, the central banks of the United States and Europe injected liquidity and aggressively lowered interest rates to prevent further spillover and contagion of the systemic impacts to the global economy. Despite a sluggish and multispeed recovery, the US economy is now gaining momentum, with signs of recovery also emerging in Europe. The GFC also raised awareness that the central bank cannot merely apply monetary policy to achieve price stability and support sustainable economic growth, but also the importance of monetary policy for financial system stability. The central bank function as lender of last resort (LOLR) again saved the financial system from heightened systemic risk.

The overarching goal of the central bank is to support sustainable economic growth through price (exchange rate) stability and financial system stability has been a solid early warning indicator for the past three centuries. Since its inception, for instance, the Bank of England has not only provided government debt and discounted loans on commercial papers to commercial banks, but also strived to prevent panic in the financial system as LOLR. Nonetheless, economic conditions, academic thinking, and political pressures have swung the pendulum of central bank focus over time. After World War II, most central banks fell under the auspices of government control to stimulate economic growth. Nevertheless, during the two decades that preceded the GFC, the policy pendulum of the central bank had swung toward emphasizing price stability through short-term interest rate instruments, namely application of the Inflation Targeting Framework (ITF). Central bank success in terms of catalyzing comparatively robust economic growth with low inflation and interest rates for two decades under ITF concealed the gradual build-up of risk and financial system fragilities that ultimately precipitated the GFC.

Consequently, in the wake of the GFC, central banks were forced back to their original mandates, namely price (or exchange rate) stability and financial system stability. As the core purpose of the central bank, price stability can support stronger economic growth with maintained macroeconomic stability. Furthermore,

numerous central banks opined that a focus on inflation control through monetary policy instruments such as ITF were no longer appropriate. In addition, the central bank must seek to maintain financial system stability in conjunction with the government and other relevant authorities. Crisis prevention is better than crisis resolution, of which the impact is more severe, costs more expensive, and management more complex. Ultimately, in the event of a crisis, the role of the central bank to stabilize and recover the economy is also vital. The central bank can actively seek financial stability through macroprudential regulation and supervision, while instituting monetary policy to lean against risk accumulation, and through payment system regulation and supervision. Moreover, the central bank also needs to tighten coordination with the government in the context of macroeconomic policy and structural reforms to support economic growth and macroeconomic stability. Similarly, central bank coordination with the government and relevant authorities, through the supervision of financial services and provision of deposit insurance, is also required to maintain financial system stability.

This chapter discusses the evolving role, policy reform, and institutional aspects of the central bank. Following this introduction, the discussion is divided into three main sections. *First*, central bank evolution since central banking was established until the present day is presented as a series of important episodes in terms of price stability and financial system stability, namely the gold standard and era of the real bills doctrine, central banking under government control, the era of central bank independence and inflation targeting and, finally, central banks in the post-crisis landscape after 2008/2009. *Second*, central bank policy reform is discussed from the perspective of several central issues, namely: monetary policy, price and exchange rate stability, foreign capital management and exchange rate stability, monetary stability and financial system stability, as well as payment system stability. *Third*, the various central bank institutional reforms, which have gained impetus over time, especially since the 1980s, are discussed, covering: credibility, independence and accountability; transparency and communication; as well as institutional arrangements and coordination with other authorities. Before closing the chapter, several strategic issues on central bank theory and policy that are currently debated and practiced at central banks in different jurisdictions around the world are presented in the final section as the foundation for further discussion in the subsequent chapters of this book.

2.2. The Evolving Role of the Central Bank

Central banking can be traced back to the seventeenth century, with the establishment of the Swedish Riksbank in 1668 as a private joint stock bank that functioned to provide loans to the government and as a clearing institution for financial transactions (Bordo, 2007). The Bank of England was subsequently established in 1694 to provide government debt and rediscounted loans through banks on commercial papers issued for trade, as well as to provide short-term liquidity facilities to banks, which later became known as LOLR (Goodhart, 2010). During the ensuing period in Europe, a number of central banks were established to finance ongoing conflicts and war. The Banque de France, for example, was established

by Napoleon in 1800 to stabilize the local currency in the wake of hyperinflation due to excessive printing of currency during the French Revolution. Another central banking model is to print and circulate money, known as a deposit bank, per the Wisselbank Amsterdam at the beginning of the 1800s (Ugolini, 2011).

The Federal Reserve System (the Fed) in the United States is a central banking system that was replicated in several countries at the turn of the twentieth century (Watchel, 2010), primarily to consolidate various payment instruments (currency) into a national legal tender and to realize financial stability. The Fed was created in 1913 to replace two preceding central banks, namely the Bank of the United States (1791–1811 and 1816–1836). The Fed was mandated with providing uniform and elastic currency to accommodate seasonal, cyclical, and secular economic shifts, while also playing the role of LOLR to ensure financial system stability (Bordo, 2007). Of note, at their inception, central banks in developing countries were established to circulate currency by consolidating the existing payment instruments or issuing new currency as the legal tender, or as a prerequisite to creating a newly independent country. Similar conditions were found in Indonesia with the establishment of BI from the nationalization of De Javasche Bank, which assumed the function of a circulating bank, succeeding Unit IV of Bank Negara Indonesia (BNI) in July 1953.

The discussion, thus far, has shown the evolution of central banks from circulating banks to the functions more commonly known and expected today. In general, central banks have been given a monopoly to print and circulate currency as the legal tender of a country. With the opening of commercial bank accounts, however, the central bank began to function as the banks' bank, facilitating all transactions in the banking system, including clearing and settlement through commercial bank accounts held at the central bank. In addition to the payment system function, a central bank also undertakes monetary operations, like the commercial papers rediscounted by the Bank of England, which is now a function of monetary policy. Furthermore, as the banks' bank, the central bank also functions as LOLR for banks facing liquidity problems, as an integral part of financial system stability. At their establishment, however, central banks were also responsible for government fiscal financing, either to finance war or purchase government debt, a function that is now restricted or even forbidden due to the adverse impact on inflation and economic stability.

In general, the central bank is the authority responsible for policies that affect money supply and credit in a country. Although formulated differently in central bank regulations per jurisdiction and discussed in academia, the following four aspects and goals are regularly found in the history of central banking (Blinder, 2010; Bordo, 2007; Goodhart, 2010). *First and foremost* is achieving price (or exchange rate) stability in line with the monetary policy adopted, such as the gold standard, fixed exchange rate, or inflation targeting regime. *Second* is regulating and supervising the payment system, including the printing and distribution of currency, regulating clearing and interbank settlement, or regulating payment instruments. *Third* is maintaining financial system stability. In the early days of central banks, this function was linked to LOLR, but nowadays also covers macroprudential regulation and supervision. In contrast, the regulation and supervision

of individual banks (microprudential) can be performed by the central bank or transferred to a financial services authority. *Fourth* is stabilizing the real economy, which typically implies job creation and robust economic growth. Initially, this function was linked to the buying of government debt or even financing the fiscal deficit, but then evolved into macroeconomic policy coordination and structural reforms with the government.

Where the pendulum swings between the main goals of the central bank evolve over time, influenced by economic conditions, political pressures, and academic thinking. Wherever the pendulum swings, however, stability will always be associated with the central bank, be it price (or exchange rate) stability, payment system stability, financial system stability, or real economic stability. In this context, Goodhart (2010) dived the focus and goals of central banking, since its inception, into three episodes: (1) the gold standard era from the 1840s to 1914, when price stability and financial stability were relatively well maintained; (2) the decades of central bank control under the auspices of the government from the 1940s to the end of the 1970s, with a focus on economic growth but prevalent economic instability; and (3) the period of central bank independence, with a focus on price stability (inflation targeting) and market mechanisms from the 1980s to 2007, when risks gradually accumulated in the financial system. The years in-between the three episodes were marred by economic distress or financial crises, including World Wars I and II from the 1920s to 1930s, the oil and government debt crises in the 1970s and the GFC. During normal times, “stability” reflected a focus on central bank policy to ensure monetary and financial system stability and, therefore, support sustainable economic growth. Meanwhile, during financial and economic crisis episodes, the term “stability” forced the central bank’s role in terms of stabilization, restructuring, and economic recovery.

2.2.1. Gold Standard Era and Real Bills Doctrine: 1840–1930

The salient developments of central bank evolution during the gold standard were maintained price and exchange rate stability, coupled with financial system stability. At their inception, central banks in various countries, including the Fed, were established to manage the gold standard, which was the prevailing international monetary system at that time. Congruent with the gold standard that remained in place until 1914, the value of a currency was defined in terms of gold and weighted according to the amount of gold reserves held. Central banks maintained large gold reserves to ensure their respective currency could be converted into gold. When gold reserves decreased due to a balance of payments (BOP) deficit or deteriorating domestic economic conditions, however, central banks were forced to raise the discount rate on their loans to commercial banks. Higher domestic interest rates attracted foreign investors and, therefore, brought in more gold to bolster central bank reserve assets.

During the gold standard era, central banks complied with the rules to maintain convertibility of their respective currency to gold over all other considerations. Consequently, gold convertibility anchored economic stability and central bank credibility. This meant the amount of currency that could be created by the central

bank was restricted by the value of gold reserve assets held and growth of money supply and, therefore, price stability was maintained. Furthermore, inflation was pegged to gold, the long-term price of which was set by market mechanisms, so price expectations were also linked to the gold price. In other words, under the gold standard, central banks were avowedly committed to price stability. The gold standard also supported financial system stability, in addition to the central bank's role as LOLR. Such conditions were supported by consistent limitations on total money supply in order to manage convertibility of the local currency into gold. Monetary policy with the gold standard also linked fund mobilization and lending by commercial banks to public income and real economic conditions.

Nonetheless, monetary policy was constrained by the stringent rules of the gold standard; therefore central banks committed instruments to stimulate economic growth. In the event of an external shock, for instance an export decline that decreased (increased) the BOP surplus (deficit), the central bank would reduce money supply to maintain currency convertibility to the gold reserves. A tighter monetary policy stance would, therefore, undermine bank credit and real economic activities, including economic growth and the labor force. Similarly, a productivity shock that reduced aggregate supply at home would require tighter monetary policy to reduce aggregate demand and avoid exacerbating the BOP, which would disrupt convertibility to the gold standard. Consequently, under the gold standard, central banks focused solely on price stability and financial system stability, without considering the underlying impacts on overall economic stability and growth.

A hot issue that is often debated is how to reconcile the gold standard while maintaining financial system stability, particularly under distress or panic conditions. As mentioned previously, monetary policy under the gold standard was achieved by regulating money supply to maintain currency convertibility to gold reserves. Operationally, however, this was achieved through the real bills doctrine, where the central bank would only purchase commercial paper from real businesses engaged in real productive activities (the real economy). Consequently, money supply was directly linked to real output/trade, therefore, monetary policy could create price stability. Similarly, exchange rate stability was also maintained because of the central bank's monetary policy response to maintain currency convertibility in line with the gold reserves held. In fact, because money supply was linked directly to credit volume and real output/trade, the financial and business cycles moved in tandem without excessive credit expansion or asset price bubbles, thus supporting financial system stability. The central bank's role in financial stability was also achieved through the function as LOLR. Per Bagehot's Rule, the central bank provided short-term liquidity facilities to solvent but illiquid banks against readily convertible securities at higher rates than market interest rates.

Problems surfaced, however, with the issuance of speculative securities or finance bills to purchase assets, particularly on the capital market. Consequently, the central bank's monetary operations generally avoided purchasing finance bills to prevent a build-up of asset price bubbles that eventually burst, leaving an economic bust in their wake, including high inflationary pressures and financial system instability (Goodhart, 2010). Such conditions occurred in the United States during the 1920s and 1930s, when the Fed, in accordance with the real bills

doctrine, refused to buy finance bills, which were proliferating in line with the capital market boom, and prohibited banks from lending to purchase stocks. In other words, the Fed tightened monetary policy in 1928 to offset the Wall Street boom, which subsequently triggered a recession and the stock market crash in 1929 (Bordo, 2007). Another issue is the central bank buying government debt securities to finance war or stimulate economic growth and create job opportunities. One reason why the Fed objected to expansive monetary operations through the purchase of government debt during the Great Depression of the 1930s was due to inconsistency with the real bills doctrine (Meltzer, 2003).¹

Although the gold standard was abandoned in 1914 and the real bills doctrine in the 1930s, several invaluable lessons also apply to central banks today. In general, both regimes produced a framework that integrates central banking theory and practices between monetary policy (price stability) and financial system stability (Goodhart, 2010). A significant aspect is the importance of central bank policy credibility, supported by the gold standard and real bills doctrine, which led to price stability and financial system stability. In the present-day context, such credibility is achievable through the ITF framework and Taylor rule to determine the appropriate interest rate policy response to achieve the inflation target. With a specific real interest rate based on the Taylor rule, the aggregate amount of real currency circulating in the economy is in line with real output, although the distribution of term deposits and bank loans is not always aligned to the real bills doctrine.

In terms of financial system stability, per the real bills doctrine, while the discounted rate and bank credit are always pegged to real output/trade, or real bills, and not to capital market speculation or government debt, or finance bills, the goals of price stability and financial system stability can both be achieved simultaneously. This cannot be guaranteed in terms of monetary policy with ITF because fundamentally, setting the policy rate does not question how and where term deposits and bank credit are intermediated. Nevertheless, the real bills doctrine and inflation targeting face the same problem, namely financial product innovation outside the banking system that leads to asset price bubbles and, therefore, financial system instability risk. In this case, production innovation occurs outside the control of money supply and credit per the real bills doctrine, while according to inflation targeting, low short-term interest rates are often the drivers of product innovation. Consequently, the central bank requires other instruments to maintain financial stability, perhaps by leaning against the wind and/or macroprudential policy regulation and supervision.

2.2.2. Central Banks under Government Control: 1940–1970

During the two World Wars, the gold standard was overburdened with government demand for financing, with central banks returning to their inaugural function of

¹It is important to note that the real bills doctrine is extremely different to thinking today, which advocates central bank monetary operations by prioritizing the purchase of government debt securities.

buying government debt. Consequently, a combination of hyperinflation, recession, and exchange rate instability threatened the international monetary system. Efforts by several countries to save the gold standard failed. For countries lacking gold, such as the United Kingdom, a return to the gold standard would mean a tighter monetary stance to overcome hyperinflation, thus exacerbating already high unemployment. Through observations of labor market rigidity, Keynes recommended utilizing monetary policy to stabilize output. Oppositely, consistent with the real bills doctrine, the United States was forced to sterilize gold inflows by tightening monetary policy due to excessive growth of money supply, credit, and stock prices. The central banks' failed commitment to the gold standard as the Great Depression descended in 1929 led to a deep economic depression in the United States and Europe, coupled with the default and closure of thousands of US banks.

Consequently, as is often the case when a crisis strikes, central banks were blamed. In many countries, including the United States and United Kingdom, the monetary policy function of the central bank was transferred to the government's control and the financial system was subjected to strict regulation (Goodhart, 2010; Group of Thirty, 2015). Under the auspices of the government, monetary policy was used as an instrument to finance the fiscal sector and stimulate economic recovery after World War II. Consequently, inflation began to soar again despite a stronger economic recovery. Meanwhile, to avoid excessive growth of money supply, the government resorted to financial repression through interest rate policy and bank credit rationing. At the same time, deposit insurance and financial safety nets were created to reduce the number of defaulting banks and incidences of financial instability from the 1940s to 1980s. Of note, domestic financial repression marked the beginning of international banking operations, including the rapid proliferation of Eurodollar securities issued in London as well as bonds issued in Latin America.

The monetary function was returned to the central banks in the 1950s, with a focus on price stability, which had been a salient issue in the United States and Europe around that time. The move was also consistent with the Bretton Woods system, agreed in 1944, to apply an adjustable fixed exchange rate system. The United States, as holder of the largest gold reserves, was the main proponent of the Bretton Woods system and the system, therefore, became synonymous with the US dollar exchange rate standard. Nonetheless, driven by the Phillips curve, which demonstrated that unemployment can be reduced only by small increases in inflation, the United States became inconsistent in its commitment to inflation, applying loose monetary policy to overcome unemployment. Other countries, therefore, such as Germany, which consistently maintained inflation by targeting money supply, began to protest US actions by threatening to exchange the dollar for gold. Consequently, the Bretton Woods system was formally abandoned in 1971, when President Nixon closed the gold window.

The collapse of the Bretton Woods system was accompanied by heightened inflationary pressures and depression in several countries at the beginning of 1974. Freed from their international commitments, central banks focused monetary policy on reducing the spike in unemployment attributable to the oil price crisis in 1973. Nevertheless, inflation expectations accelerated rapidly, hence compelling

the central banks to apply monetary targeting, usurping unemployment. In reality, the strategy to target money supply was unable to control inflation due to, among others, unstable demand in line with financial sector development and foreign capital flows. In 1979, when the second oil price shock occurred, most central banks began to focus on inflation control rather than just leaning against the wind to anticipate bumps in unemployment. Another important aspect of the second oil price shock was the expansion of banking activities through international financial transactions to recycle oil proceeds from net producers to net importer EME. Several EMEs, particularly in Latin America during the 1970s, issued government bonds and/or took out large loans from United States and European banks, which subsequently triggered another crisis when interest rates increased. In advanced countries, the issue of financial stability was also raised among central banks during the middle of the 1970s due to several defaulting banks operating internationally, including the Franklin National Bank in the United States and Bank Herstatt in Germany in 1974.²

2.2.3. Central Bank Independence and Inflation Targeting: 1980–2007

High inflation compelled many central banks to focus more on price stability. Initially, monetary policy was directed toward controlling inflation through monetary targeting. Nevertheless, controlled inflation was unable to stimulate growth and, therefore, central banks in advanced and EME sought to bring inflation down to low levels. A new monetary policy framework, known as the ITF, which was introduced by New Zealand in 1988, became increasingly prominent. Numerous central banks around the world applied various iterations of ITF, from rigid to flexible, and with different weights for achieving the inflation target with a trade-off for economic growth (output gap). New Zealand was one example where a rigid ITF was applied, while the US Federal Reserve, European Central Bank (ECB), and Bank of Japan (BOJ) favored a more flexible approach that also considered economic growth and employment on top of the inflation target.

Several characteristics of ITF supported central bank monetary policy commitment and credibility in terms of price stability (Bernanke, Gertler, & Gilchrist, 1999; King, 2004; Svenson, 1999). ITF shared a number of common aspects with central bank policy during the gold standard and real bills doctrine. *First*, commitment to achieve a published inflation target. Success in terms of

²The need for central banks to maintain financial system stability in addition to price stability subsequently nurtured international cooperation through establishment of the Bank for International Settlements (BIS), which seeks to strengthen capital and bank supervision standards through the Basel Committee on Banking Supervision (BCBS) and the payment system through the Committee on Payments and Settlement Systems (CPSS). The BIS committees also reinforced two other multinational institutions, namely the International Monetary Fund (IMF), which promotes international monetary cooperation, and the World Bank, which advocates cooperation and financial development.

achieving the inflation target would anchor future inflation expectations. Per the gold standard, commitment to the inflation target was achieved through currency convertibility to the gold reserves held by the central bank. *Second*, monetary policy consistency to achieve the inflation target, by setting interest rates (for instance using the Taylor rule) to achieve the inflation target two years ahead. Under the gold standard, consistency was supported through monetary targeting in line with the reserve assets held to ensure exchange rate convertibility. *Third*, central bank independence in terms of monetary policy to achieve the inflation target. This is analogous with the central bank's mandate to maintain currency convertibility under the gold standard. *Fourth*, central bank transparency and communication concerning inflation, macroeconomic projections, and even monetary policy direction. This aspect was strengthened under ITF in order to achieve strong central bank credibility, similar to that during the gold standard era.

ITF, as a monetary policy framework, was considered a success when applied in most advanced countries and many EME (Berg, 2013; Taylor, 2014). Inflation and volatility decreased sharply, while interest rates also fell. Moreover, after the deep recessions felt in the 1970s, downturns in the 1980s and thereafter became less frequent and shallower (including in the United States). In fact, economic growth continued to accelerate, driving the rapid expansion of trade and investment between countries. The subsequent economic boom, coupled with low inflation, robust economic growth, lower unemployment and low interest rates, endured for the next two decades during an era that became known as the Great Moderation.

Nevertheless, financial instability simultaneously returned. High credit growth, often linked to rapid property and housing development, was considered as the root cause. Robust economic growth, together with low inflation and interest rates prompted a domestic and global economic boom. Widespread financial deregulation in the 1980s further accelerated finance in the boom economy. In the United States, for example, removal of restrictions on deposit rates at financial institutions (savings and loans) in the 1980s accelerated short-term funding for long-term housing loans. Consequently, when policy rates were raised in the 1990s, savings and loans institutions were sent into bankruptcy. Furthermore, a US stock price shock in 1987 quickly spread to other countries. A housing credit and banking crisis also befell Japan at the beginning of the 1990s. In addition, several banks extending international loans, primarily to Latin America, experienced default in the 1980s, which again threatened financial instability. The subsequent boom in Asia, known as the East Asian Miracle, attracted an influx of foreign capital flows, including private external debt, which precipitated the Asian Financial Crisis in 1997/1998.

Credit growth in the US private sector, however, particularly the housing sector, continued to outpace bank deposit funding for more than two decades (Schularick & Taylor, 2009). Financial innovation to meet the credit demand boom changed the banks' business model as follows: (1) selling government debt assets (treasury bills and bonds); (2) applying "originate to distribute" strategy with credit securitization, particularly mortgage facilities, and selling the products to nonbank financial institutions and the capital market; and (3) relying on wholesale funding by issuing bonds and/or credit securitization (Goodhart, 2010). This change fostered greater risk-taking behavior and financial system fragilities.

Furthermore, during the economic boom and period of asset price bubbles, banks increased leverage through excessive credit growth based on rising asset prices.

Although banks in the United States and Europe were subject to the capital requirements of Basel II, leverage was still increased by expanding asset portfolios (AAA) with mortgage-backed securities (MBS). Financial procyclicality increased due to risk-weighted assets and mark-to-market accounting. During the boom period, risk-weighted assets tended to be too low, while mark-to-market accounting expanded the capital base. Financial innovation, coupled with high leverage and risk-taking behavior during the two decades of economic boom ultimately triggered the GFC, the most severe crisis since the Great Depression, from September 2008 to March 2009, the fallout of which was felt worldwide.

2.2.4. Central Banks and the Global Financial Crisis of 2008/2009

The GFC of 2008/2009 forced central banks to stabilize the financial system again and rescue the economy. This time, however, the necessary central banking actions extended beyond LOLR. Central banks in advanced countries were also forced into post-crisis financial stabilization measures by expanding liquidity facilities to nonbank financial institutions and the financial markets, while also instituting accommodative monetary policy to save the economy. The US Fed, for example, issued four new financial stabilization instruments, namely the Term Auction Facility (TAF) to provide short-term liquidity using (1) securities issued by banks and nonbank financial institutions as collateral between December 2007 and April 2009; (2) loans to facilitate the purchase of Bear Stearns by JP Morgan; and (3) support for American International Group (AIG) and jurisdiction to repay interest on the statutory reserve requirement (Goodfriend, 2010). Consequently, the balance of Federal Reserve assets skyrocketed, from USD 0.9 trillion in July 2007 to USD 2.8 trillion in July 2011. The Fed subsequently adopted quantitative easing (QE) to recover the economy from crisis, prompting Fed assets to climb further to USD 4.2 trillion in April 2014 (Mehrling, 2015).

The exorbitant cost of crisis recovery, combined with the far-reaching fallout, reaffirmed that the central bank mandate of monetary policy alone was insufficient, financial system stability also had to be maintained. In fact, the mandate to maintain financial system stability was present since the inception of central banks in the seventeenth century, namely through the central bank acting as the banks' bank under normal conditions and functioning as LOLR under crisis conditions. In the present context, the severe impact and high cost of the GFC in 2008/2009 raised awareness that financial system stability is desirable more to prevent a crisis rather than overcome a crisis. The financial stability function of a central bank must be combined with monetary policy that has evolved to require sound credibility, similar to conditions under the gold standard and ITF eras.

Nonetheless, central bank evolution in terms of monetary policy and financial system stability is heavily influenced by prevailing economic conditions, political pressure, and nascent theories. Oftentimes, the important role of central banks is forgotten or even ignored during economic boom periods, but when a crisis strikes, economic stabilization and recovery become crucial. That attitude toward central

banks needs to stop. Many important lessons have been learned from the evolution of central banking since its inception until the present day, while the GFC in 2008/2009 greatly reinforced the various efforts required moving forward as follows.

First, price (or exchange rate) stability must remain the core mandate of the central bank. From past experience, the government's exploitation of the central bank printing money to finance war, buy debt, or for post-war economic recovery left an indelible mark on the community through high inflation, low growth and economic uncertainty. On the other hand, central bank commitment and credibility to maintain currency convertibility during the gold standard era, and to control money supply in accordance with real output during the real bills doctrine, ensured price stability, business assurance, and economic growth. Likewise, central bank credibility through ITF in the decades since the 1980s helped to lower inflation and interest rates, while fostering robust economic growth during the era of Great Moderation. The need to anchor inflation in the future is also supported by a synthesis of Keynesian and Neoclassic theories, namely that money in the near term only affects prices but, in the long term, there is a trade-off between inflation and economic growth.

The GFC of 2008/2009 clearly showed that low inflation due to monetary policy credibility could not guarantee macroeconomic and financial system stability. Prevailing thinking prior to the GFC, namely that if a central bank could stabilize inflation in the near term (e.g., up to two years, such as ITF) the economy would return itself to a steady state in the long term, proved to be erroneous (Borio, 2011). That perspective formed the basis of macroeconomic models at the time, stating the price rigidity was the only cause of economic deviation from its steady state (Walsh, 2010; Woodford, 2003). Conversely, the GFC also showed that a period of robust economic growth, coupled with low inflation and interest rates, due to successful monetary policy, actually creates financial system instability. Financial innovation and risk-taking behavior led to high leverage along with excessive credit growth and asset price bubbles, which ultimately resulted in the severe GFC of 2008/2009.

Second, central banks must reinforce their mandates in terms of monetary policy, financial system stability, and the payment system, similar to when central banks were first established. As highlighted previously, the acute fallout and high cost of the GFC provided valuable insight that crisis prevention is better than crisis resolution. Experience of the GFC in 2008/2009 reversed the previously held view that the monetary policy function should be separate from financial system stability (Borio, 2010). Prevailing thought before the GFC stated that: (1) monetary policy should focus on price stability; (2) microprudential regulation and supervision, with a focus on individual financial institutions can and is sufficient to maintain overall financial system stability; (3) financial institution regulation and supervision should be separate from the central bank to avoid conflicts of interest; and (4) the central bank's role as LOLR is adequate to ensure financial stability. The GFC decisively dispelled those viewpoints. Financial instability actually originated from financial system and macroeconomic linkages through intermediation. The crisis transpired because the financial cycle was bombarded by excessive credit growth, debt accumulation or high leverage, and uncontrolled

asset price bubbles. Therefore, the role of the central bank to maintain financial system stability and prevent crises must be strengthened through macroprudential regulation and supervision that focuses on a macrofinancial perspective and systemic risk (Borio, 2010).

Moreover, the GFC of 2008/2009 also clarified that crisis resolution through low interest rates and large-scale monetary injections is not always effective and requires a protracted recovery period. Central banks in advanced countries must make every effort to expand their balance sheets in order to ensure healthy long-term interest rates and financial conditions when instituting ultra-loose monetary policy (Borio & Disyatat, 2010). The experience of the GFC reversed previously held beliefs, including ITF, that short-term interest rates would be sufficient to transmit the effect of monetary policy to the economy. Monetary policy, therefore, no longer focuses merely on controlling short-term interest rates, while simultaneously anchoring future expectations, but on setting the full interest rate structure for all subsequent tenors (Svensson, 2003). Prevailing view before the GFC even denied the possibility that central banks, such as the Fed, ECB, and BOJ, would lower nominal interest rates to 0% (Ahearne, Gagnon, Haltmaier, & Kamin, 2002; Orphanides & Wieland, 1998). The GFC also showed that financial frictions occur because of financial innovation, asymmetric information, valuation methods and price setting, and even risk-taking behavior, which prompted a fundamental change when determining interest rates and the monetary policy transmission mechanisms to the economy.

Third, the GFC of 2008/2009 showed how rapidly spillover can propagate from central bank policy in one country to another and the large impacts thereof. Liquidity injections by central banks in advanced countries triggered massive foreign capital flows and high exchange rate appreciation in EME. Nonetheless, the normalization process of monetary policy in the United States since May 2013 and monetary policy divergence from the ECB and BOJ, has triggered widespread uncertainty in the global financial system, affecting foreign capital flow volatility and exchange rates in EME. The GFC demonstrated that monetary policy is becoming increasingly globalized. Despite not undermining its ability to control conditions on domestic financial markets and the national economy, the Fed now pays more attention to the global aspects underlying monetary policymaking (Bernanke, 2007). Financial interconnectedness between countries in the monetary and international financial system has also tended to amplify the accumulation of uncertainty (Borio, 2014), while flows of foreign capital led to excessive credit growth that subsequently triggered financial instability (Borio, McCauley, & McGuire, 2011). In fact, as external debt spiraled, along with USD denominated bonds issued by EME, including Asia, the impact of global spillovers from the Fed's policy to economic and financial conditions in EMEs accelerated and increased in magnitude (McCauley, McGuire, & Sushko, 2015).

The GFC experience has reversed the former doctrine of "keep-your-house-in-order," namely that if each central bank acted in accordance with their own respective economy, then global economic and monetary conditions would also be sound. This kind of doctrine was analogous with the view that if regulation and supervision could guarantee the soundness of individual financial institutions,

then financial system stability would also be maintained. This doctrine advocated that each central bank should focus on price stability in their respective domestic economy, through ITF application for instance, and follow a free-floating exchange rate regime (Rose, 2007). In many ways, such a viewpoint assumed that the international financial and monetary system worked efficiently and each financial asset was a perfect substitute, thus the law of one price and uncovered interest parity applied. In reality, however, the GFC demonstrated that such a viewpoint is not always correct and, therefore, many central banks, particularly in EME, must engage in exchange rate stabilization and foreign capital flow management to protect the domestic economy from the impact of global spillovers.

Fourth, the evolving role of central banks, from their inception until the GFC in 2008/2009, has shown that central banks must be free from political intervention if they are to achieve price stability and financial system stability. Past experience from when the central bank was under government control, whether to finance war or purchase government debt, always resulted in high inflation and subsequent recession. The fiscal domination of central bank policy must be avoided. Nevertheless, central bank independence is never absolute. The central bank must remain avowedly committed to its mandate in order to remain credible, including in terms of developing a sound policy framework, strong leadership, and solid capabilities. The central bank must act to strengthen accountability and transparency through periodic reports, testimony in front of parliament, as well as communicating policy to the public and stakeholders. Moreover, independence does not absolve the central bank from building policy coordination with the Government and other relevant authorities. In contrast, central bank independence should be founded on macroeconomic and financial system policy interdependence as well as economic structural reforms to achieve robust economic growth and create broad job opportunities, supported by maintained macroeconomic and financial system stability.

2.3. Central Bank Policy Reform

Coordination between monetary, macroprudential, and payment system policy with fiscal policy and government structural reforms was desperately required to accelerate the economic recovery in the wake of the GFC in 2008/2009. Hitherto, the onus for the GFC resolution measures and subsequent economic recovery have been placed on the central banks with monetary policy and financial stability; hence, El-Erian (2016) emphasized the difficult task of the central bank as “the only game in town.” Nonetheless, although the central banks worked diligently and successfully saved the financial system from total failure and economic collapse and from a protracted depression state, such as the Great Depression of the 1930s, the economies of advanced countries and the global economy have grown sluggishly against a backdrop of ubiquitous uncertainty blighting global financial markets. The role of the central bank must be strengthened further in order to stimulate robust economic growth and ensure lasting financial stability. That is still not enough, however. Comprehensive policy concerning structural reforms to more seriously nurture inclusive economic growth is required, along

with fiscal policy to stimulate demand, overcoming the debt overhang, as well as regional and international institutional arrangements (El-Erian, 2016).

Several schools of thought have been proposed to strengthen modern central bank policy after the GFC of 2008/2009 (Bank for International Settlements [BIS], 2011; Borio, 2011; Committee on International Economic Policy and Reform, 2011; Georgsson, Vredin, & Sommar, 2015; Group of Thirty, 2015; Orphanides, 2011). Strengthening central bank policy encompasses several aspects to support sustainable economic growth through monetary and macroprudential policy, foreign capital flow management and payment system policy. Central bank evolution has provided a number of valuable lessons, while experience from before and after the GFC of 2008/2009 represents an important foundation to rebuild modern central banks.

2.3.1. Monetary Policy, Price Stability, and Exchange Rates

Central bank evolution has shown that policy credibility is vital to create and maintain stability in the economy. During the gold standard era, commitment to currency convertibility in line with the gold reserves held combined three aspects of stability achieved by the central bank, namely; (1) exchange rate stability in line with the currency's convertibility, (2) financial system stability because total money supply (and credit) was in line with real demand and the reserve assets, and (3) price stability due to currency convertibility and controlled money supply. During the ITF era, price stability was achieved through central bank commitment to the inflation target, with consistent short-term interest rates. Notwithstanding, exchange rate stability and growth of money supply (and credit) depended heavily on market mechanisms and behavior in the financial system. Under both regimes, economic growth is considered the second goal after price stability.

The GFC of 2008/2009 exposed weakness in the rigid application of ITF and, therefore, many countries opted for a flexible ITF. Two issues emerged in its implementation, however: (1) the policy response to heightened exchange rate volatility, and (2) the accumulation of asset price bubbles, such as a stock price boom or property price boom. Regarding the exchange rate, central banks face a policy trilemma concerning interest rate policy efficacy in terms of achieving price stability with exchange rate volatility and foreign capital flows. In that context, several studies have shown how a central bank can overcome the policy trilemma. Calvo and Reinhart (2002), for example, pointed out the phenomenon of "fear of floating" in the case of exchange rate policy in various countries as more embraced an intermediate system rather than the hard pegged or purely floating systems. Meanwhile, empirical studies show that EMEs applying ITF employ a strategy where the central bank responds well to inflation and real exchange rate when setting the monetary policy rate (Aizenman, Hutchison, & Noy, 2011; Mohanty & Klau, 2004).

The importance of exchange rate stabilization compelled several economists, including Ostry, Ghosh, and Chamon (2012a, 2012b), to recommend a dual-target regime, namely exchange rate stabilization to support attainment of the inflation target, in order to bolster central bank monetary policy credibility. The prerequisite,

while predicated on ITF, is that the exchange rate target is used to mitigate the undesirable impacts of capital flow shocks on inflation expectations, through the direct influence of exchange rate pass-through or indirectly through domestic demand. With the effect of exchange rate fluctuations included in the assessment and projections of the central bank, exchange rate targeting is directly more effective than an interest rate response to the impact of exchange rates on inflation. Ostry et al. (2012a, 2012b) recommended that central banks in EMEs apply the dual-target regime. Furthermore, central banks in EME also have a secondary instrument at their disposal in addition to the interest rate, namely foreign exchange intervention. The latest study by Blanchard, Adler, and Filho (2015) investigated the effectiveness of foreign exchange intervention as an instrument to mitigate exchange rate pressures originating from foreign capital flows. In other words, monetary policy with dual targets, namely the inflation target and exchange rate stabilization, using two instruments, namely interest rates and foreign exchange intervention, would increase, not decrease, central bank credibility.

The latest empirical evidence after the GFC in 2008/2009 demonstrated the importance of central bank policy in response to asset price bubbles, particularly property prices. Jorda, Schularick, and Taylor (2014), for example, provided important evidence on the correlation between monetary conditions, house prices, and credit growth from historical data stretching back 140 years from 14 advanced countries. The discussion focused on how monetary policy could respond to the accumulation of risk, reflected in procyclical housing bubbles and credit booms. This issue also touches upon another debate, namely lean versus clean: whether it would be better for the central bank to lean against the wind to prevent the bubbles from bursting, or better to wait until the bubble has burst and then clean up the mess through aggressively loose monetary policy. The “clean” view was embraced by the Federal Reserve under the leadership of Alan Greenspan, otherwise known as the Jackson Hole consensus. A number of considerations underlie that view: an investment boom driven by higher productivity, bubbles due to lower risk premiums and irrational exuberance, as well as interest rate hikes that might be ineffective against the bubbles but could actually cause the bubble to burst and spillover into the economy (Greenspan, 2002).

Nevertheless, the GFC of 2008/2009 dispelled that view after clearly showing potential risk from excessive credit and leverage, driven by the bubbles. Consequently, a leaning monetary policy was considered more appropriate to avoid bubbles, rather than cleaning up the fallout of a crisis. Experience in Australia, for example, showed that leaning monetary policy can be successfully implemented. In response to excessive credit growth in the housing sector in 2002 and 2003, the Reserve Bank of Australia gradually raised interest rates despite the inflation outlook remaining under control within the target (Bloxham, Kent, & Robson, 2011). Consensus now strongly supports central banks seriously considering aspects of financial stability and leaning against the wind despite financial stability not legally being part of the central bank’s mandate. Bernanke (2009), for instance, stated that the Fed played an influential role in overcoming the crisis in the United States and, therefore, its institutional ability to support financial stability and drive economic recovery with low inflation must be maintained.

2.3.2. Exchange Rate Stability and Foreign Capital Flows

As mentioned previously, not only did exchange rate volatility increase after the GFC, but also the volatility of foreign capital flows, which undermined the effectiveness of central bank policy to accomplish the domestic goals in line with the policy trilemma of an open economy. Exchange rate stabilization through foreign exchange intervention constitutes one option available to the central bank. Nonetheless, the magnitude of foreign capital flow volatility undermined the efficacy of the central bank's interest rate policy and foreign exchange intervention, which also incurred considerable expense. In fact, exchange rate and foreign capital flow volatility can trigger financial imbalances and thus prompt macroeconomic and financial system instability (Borio & Disyatat, 2011). Consequently, the regulation of foreign capital flows represents another policy option available to the modern central bank (Eichengreen et al., 2011).

Several follow-up studies show the importance of foreign capital flow management to help overcome the policy trilemma faced. Klein and Shambaugh (2013) investigated how far foreign capital flow management policy, through temporary and directed capital controls, or limited exchange rate flexibility, could realize full autonomy in terms of central bank policy. The findings showed that foreign capital flow management under a flexible exchange rate regime could improve monetary policy autonomy. Nevertheless, capital flow management under a fixed exchange rate regime would be less advantageous to monetary policy autonomy compared to full capital mobility. As has been applied worldwide, extensive and long-term capital controls would break the links between domestic and international interest rates under a fixed exchange rate system. Furthermore, inflation is generally more manageable under a fixed exchange rate regime and, therefore, the gains in terms of monetary policy autonomy are more limited.

Rey (2013) also extolled that importance of foreign capital flow management to overcome the monetary policy trilemma based on the increased comovement of capital flows, asset prices, and economic growth between countries in the global financial cycle. Consequently, Rey (2013) urged central banks to change the trilemma into a dilemma, namely strengthening monetary policy autonomy and exchange rate stabilization through direct foreign capital flow management or with macroprudential policy. The study showed that the salient determinants of the global financial cycle are monetary policy in advanced countries, which influences capital flow leverage, and credit growth in the international financial system, the impact of which is transmitted to numerous countries, especially EMEs, thus interest rate policy and exchange rate flexibility are unable to protect an economy from the global financial cycle.

Obstfeld (2015) went further, stating that the impact of financial globalization has not only created a monetary policy trilemma but also and financial trilemma. The study evaluated the ability of monetary policy in EMEs to mitigate the impact of global monetary and financial fluctuations. Congruent with the monetary policy trilemma, the study confirmed that EMEs applying a flexible exchange rate regime are better disposed toward monetary policy autonomy. Nevertheless, changes to the exchange rate regime are unable to protect the domestic economy from global monetary and financial shocks. The most important channels of

international monetary and financial policy transmission are long-term interest rates and risk premiums, while financial globalization has been shown to not only drive capital flows but also distribute risk across jurisdictions. Despite mitigating global spillovers through exchange rates, monetary policy cannot alleviate the impact of global capital flows and financial risks in the domestic financial system. Therefore, the monetary policy trilemma expanded to the financial trilemma, namely policy autonomy with the overarching influence of global financial integration while maintaining domestic financial system stability.

2.3.3. Monetary and Financial System Stability

The GFC of 2008/2009 also raised awareness concerning the need to restore the mandate of financial stability to the central bank, on top of price stability that has been the goal since the beginning. The existing definitions of financial system stability are not always exactly the same between policymakers and academia, but fundamentally refer to conditions where the financial system functions soundly in an economy and demonstrates resilience to the various shocks that could emerge (Allen & Wood, 2006; Mishkin, 1999). Mandating the central bank with financial stability would also correct the policy reformulation and institutional reforms of the previous crisis. After the Asian Financial Crisis in 1997/1998, for example, various authorities focused on the most appropriate approach for that time. Central banks focused on price (and exchange rate) stability through monetary policy. Furthermore, the regulation and supervision of financial institutions was transferred away from the central bank to newly formed financial services authorities, such as in South Korea and Indonesia. The GFC in 2008/2009, however, showed just how important it is to look at the financial system from a macrofinancial perspective, in which systemic risk must be mitigated through macroprudential policy.

The role of central banks in terms of financial stability is achieved directly through macroprudential policy and as LOLR, and indirectly through monetary and payment system policy (Haldane, 2004; Padoa-Schioppa, 2003). The central bank's mandate to support financial stability is primarily accomplished through the macroprudential regulation and supervision of financial institutions from a macrofinancial perspective and a focus on systemic risk. Macrofinancial linkages between financial institutions and the economy often exacerbate financial procyclicality through a faster acceleration during periods of economic expansion and a deeper recession during an economic downturn. Procyclicality, if not managed accordingly, will accelerate the boom-bust periods of the financial cycle, which could result in a crisis. Therefore, macroprudential policy is directed toward managing financial procyclicality, with a strong focus on credit booms and asset price bubbles, to avoid macrofinancial imbalances that often lead to financial crises. In addition, macroprudential policy is also directed toward managing interconnectedness and financial networks, particularly through the interbank money market and financial infrastructure, to avoid systemic risk. Default at a financial institution can trigger a crisis, along with macrofinancial procyclicality as well as domestic or international economic shocks, while a systemic crisis to the entire financial system can occur rapidly due to interconnectedness and financial networks.

In addition to macroprudential policy, central banks also support financial stability through monetary and payment system policies as well as foreign capital flow management. Monetary policy supports financial stability by maintaining macroeconomic stability, and the primary target is to achieve price (and exchange rate) stability. This includes regulation and supervision of the money markets and foreign exchange market as an integral part of monetary policy, which also supports financial stability in terms of mitigating systemic risk from interconnectedness and financial networks. The same considerations also apply to the payment system, for which the primary goal is to provide a secure, reliable, and efficient payment system as a fundamental part of monetary policy effectiveness. Meanwhile, managing foreign capital flows protects financial stability from external spillovers, on top of the primary mandate to support exchange rate stability and avoid sudden-stop and BOP crises.

The central bank is the most appropriate institution to conduct macroprudential regulation and supervision. The central bank has the capacity to perform macroeconomic and macrofinancial surveillance as well as the instruments to implement macroprudential policy (Kawai & Morgan, 2012). Furthermore, a BIS (2011) study of 13 advanced and emerging market countries concluded that central banks must have direct involvement in financial system stability policymaking and implementation if the policy is to be effective based on three salient explanations. *First*, the monetary policy framework and task implementation focuses the central bank on macroeconomic analysis and projections, while complementing understanding of the financial markets, institutions and infrastructure, all matters of great import in terms of macroprudential policy implementation. *Second*, financial system instability can be caused by and have a significant impact on economic performance, with fundamental consequences on economic activities, price, and exchange rate stability, as well as monetary policy transmission effectiveness. Central banks are already familiar with evaluating financial system and economic interconnectedness, or the macrofinancial aspects, as the focus of macroprudential policy. *Third*, the central bank is a source of liquidity in the financial system and economy through monetary policy and the LOLR function, and the availability of liquidity is crucial when maintaining financial stability.

2.3.4. Payment Systemic Stability

Printing and circulating currency have been the overarching purposes of the central bank since the beginning (Roberts & Velde, 2014) and, therefore, have become the basis of central bank duties in terms of monetary policy, the payment system, and financial system stability (Blinder, 2010; Goodhart, 2010). In the payment system, the central bank does not merely print currency as legal tender but also holds the accounts of the banking sector as final settlement assets for all their financial transactions, often referred to as currency outside bank or central bank money. Meanwhile, commercial banks create demand deposits by opening savings deposits for the public, otherwise known as inside money or commercial bank money. Both types of money have developed simultaneously in modern economic transactions, where public transactions are facilitated by commercial bank

money and then interbank transactions are facilitated by central bank money (BIS, 2003). Confidence in commercial bank money lies in the respective commercial bank's ability to liquidate term deposits and other obligations to the public as required from the liquid assets held, or through financial transactions with other commercial banks and/or through their account held at the central bank. On the other hand, confidence in central bank money lies in the central bank's ability to provide liquidity to commercial banks as required and maintain a stable value of the money created, namely price and exchange rate stability.

The payment system encompasses all instruments and conventions to transfer funds from one bank to another (Borio, Russo, & van de Bergh, 1992). Therefore, the payment system consists of three main elements or core processes: (1) authorization to make payments from a customer to a bank; (2) exchange payment instructions between banks, known as clearing; and (3) transaction settlement through the respective accounts of each bank at the central bank (Sheppard, 1996). Large value transactions are typically settled through the real-time gross settlement system (RTGS), while small-value transactions are settled through the national clearing system. It is important to note that the payment system also includes payment mechanisms for government bond transactions (e.g., through the BI – Scripless Securities Settlement System, BI-S4) as well as stock and bond transactions on the capital market. Considering the importance of the payment system to accelerate the volume and efficiency of various transactions in the economy, central banks around the world have adopted best practices in the form of core principles when operating and regulating the payment system (BIS, 2001).

The function of a central bank in the payment system correlates closely with its duties in monetary policy and financial stability (ECB, 2007). In general, monetary stability supports the continuity of various transactions and economic growth and, therefore, underpins financial stability and payment system reliability. Furthermore, a reliable payment system is a prerequisite to monetary policy implementation due to its support toward monetary operations and money market security and efficiency. Through monetary policy implementation, central banks set their policy rates and undertake monetary operations to influence market liquidity in line with the policy rates set. Effective monetary operations require: (1) an active interbank money market that transmits lending and borrowing between banks suffering from inadequate liquidity and banks with excess liquidity, which is also an indicator of monetary conditions, reflecting general economic activity; and (2) the central bank to predict the determinants of market liquidity, specifically the circulation of currency between the government and public, such as tax revenues and budget spending (Sheppard, 1996). A secure, reliable, and efficient payment system supports effective monetary operations and transmission to the economy in general (Gaspar & Russo, 2006).

The payment system is also inextricably linked to the financial stability function of the central bank. The financial system is said to be stable if it is resilient to shocks and distress, thus preventing disruptions to money supply and credit, the intermediation function to allocate savings deposits to various investment activities, as well as the payment process in the economy. In this case, market infrastructure, including a sound, efficient, and resilient payment system, is a prerequisite

for the markets and financial institutions to function (BIS, 2000). Conversely, financial stability is crucial to the function of the payment system and, ultimately, the success of monetary policy, both of which represent the fundamental prerequisites for sustainable economic growth.

Systemic disruptions can appear in the payment system due to, for instance, problems at a participating bank or because of technical difficulties with the system operator. Such disruptions can influence the position of liquidity at other banks, demand for currency, and ability to settle various financial instruments (Bogov, 2011). Eventually, such conditions spill over to interest rates, which subsequently impact trade activities and price setting mechanisms on the markets. Moreover, contagion risk could materialize due to the mutual transactions between banks, which hold the same positions as one another in the payment system. The inability (actual or perceived) of a bank to settle its obligations under financial market distress can undermine confidence and trigger systemic risk on the default of financial institutions. This is where the central bank plays a vital role in financial stability through monetary operations in the form of intervention when the markets fail to function as market maker of first resort, or by providing liquidity facilities to solvent banks as LOLR in accordance with Bagehot's Rule (ECB, 2007).

Over the past few decades, the payment system has developed rapidly in line with the evolution of the modern financial system (Fung, 2012). Concerning the retail payment system, payment innovation has progressed quickly, providing public access to various new payment instruments, such as the proliferation and widespread use of automated teller machines (ATM), debit and credit cards, mobile and internet banking, as well as electronic money. Central banks need to maintain the reliability and efficiency of the retail payment system, as a crucial element of a functioning economy, in addition to the influence of payment innovation on future public demand for currency. Meanwhile, concerning the large value payment system, interbank transaction settlement system security and efficiency are key to national financial system stability and resilience to shocks, such as the GFC. The large value payment systems around the world continue to function despite onerous liquidity issues in the financial system, even when faced with bankruptcies. Such conditions prove that market infrastructure resilience, including the payment system, is imperative for financial system stability and efficiency.

The GFC of 2008/2009 also precipitated a new phenomenon that prompted a fundamental change in the current and future financial system, namely financial technology, or FinTech (Horn, Geerts, & Coolen, 2014; PwC, 2026; Walport, 2015; World Economic Forum, 2015). Against a backdrop of more stringent capital and prudential regulations for financial institutions, small-medium FinTech companies have sprung up that use information technology and telecommunications in very innovative ways to offer direct financial services to their customers. The business segments include: (1) using technology to directly present financial services to consumers, thereby competing and even replacing traditional financial services, such as a competitive peer-to-peer borrowing platform with banks when lending to micro, small, and medium enterprises (MSME); and (2) using

new technology to strengthen or optimize the outreach of banking markets, hedge funds, and other financial service businesses. Financial technology (FinTech) companies enable the digitization of financial products and services, thereby extending the outreach to customers, providing alternative solutions to improve the processes and supporting administration efficiency, including new technology to visualize and analyze the data. On one hand, the rapid emergence of FinTech has extended outreach, while increasing the volume and efficiency of financial services and the payment system, particularly the retail payment system. On the other hand, however, financial innovation outside of formal financial institutions through the rapid emergence of FinTech could create financial system fragmentation and financial system stability risks that demand vigilance and prudential regulation by the central bank.

Payment system reliability and security are crucial for financial system stability, efficiency, and effectiveness. In this case, payment system failure could trigger potential financial systemic risk, particularly if the large value payment system is affected (Comotto, 2011). Payment default of an asset or transaction would: (1) cause the payment system participant to incur loss on the default payment (credit risk); and (2) exacerbate liquidity conditions at each respective payment system participant (liquidity risk). Problems at one payment system participant could inadvertently spill over to affect the customers, thereby creating a loss of confidence and squeezing market liquidity (market liquidity risk), which would propagate systemically through contagion to the rest of the financial system and even the economy. Payment system failure could originate at participating financial institutions because of inadequate liquidity or insolvency, or because payment default is caused by the issuing financial institution or due to a loss of confidence. Therefore, to maintain payment system reliability and security, central bank regulation and supervision aim to: (1) prevent a build-up of systemic risk and maintain financial stability; (2) enhance system efficiency and payment instruments; (3) safeguard security and public confidence in their payment assets; and (4) protect monetary policy transmission channels (Kokkola, 2010). In pursuance of its target, a central bank's general functions are from various aspects of the payment system, as operator, initiator, and supervisory authority.

2.4. Central Bank Institutional Reform

Central bank credibility is vital in pursuit of the goals, namely price (and exchange rate) stability, financial system stability and payment system stability. Under the gold standard, credibility was the central bank's obligation through commitment to maintain currency convertibility in line with the gold reserves held. After a brief disappearance under government control, central bank credibility was restored under ITF through institutional reform that granted central bank independence, while strengthening policy accountability and transparency. Academic research also supported the institutional reforms, while actual practices at various central banks showed the institutional reforms had enhanced central bank credibility, particularly in terms of maintaining price stability.

2.4.1. Credibility, Independence, and Accountability

As discussed in the previous section, central bank policy practices and implementation during the ITF era were characterized by two types of institutional reform. *First*, providing greater independence or jurisdiction to the central bank. *Second*, refocusing on a single mandate, namely price stability. Both steps were taken in response to low central bank credibility during the era under government control. After that time, not only were central banks redirected toward stimulating economic growth but were also given autonomy over policymaking and implementation. Theoretically, the problem of central bank credibility correlates with monetary policy time inconsistency in terms of achieving price stability (Kydland & Prescott, 1977).

Monetary policy inconsistency can occur because central banks assign larger weights to output (job opportunities) and/or formulate policy responses that are inconsistent with the inflation target (Barro & Gordon, 1983). For example, central banks initially directed monetary policy toward attaining a given inflation target. Nonetheless, political pressure to reduce interest rates and/or finance government spending through the creation of money led to inconsistent monetary policy and caused a higher inflation bias. In that context, monetary policy inconsistency is analogous to an inflation tax because of the drive toward stronger economic growth. From the central bank's perspective, policy inconsistency can be interpreted as seigniorage, thus triggering inflation.

A few measures are available to overcome policy inconsistency, including: (1) delegating authority or granting independence to the central bank; and (2) introducing policy rules (Calvo, 1978; Kydland & Prescott, 1977; Rogoff, 1985). Fundamentally, independence is delegating authority from the public through the government to the policy target of the central bank. For the central bank, however, independence is required to limit political intervention in the legal mandate. In developing countries, referring to experience in many countries as well as comparatively underdeveloped political conditions, central bank independence is required to sustainably ensure the goal of economic development remains in line with the gravitas of stability. In an ITF context, for instance, setting the inflation target to be achieved by the central bank represents the delegation of public authority pursuant to prevailing laws. The inflation target may be set by the government itself or by the government based on the central bank's recommendations. At several central banks, including New Zealand and the United Kingdom, the delegation of authority for the inflation target is contained within a remit to the central bank.

Independence must be accompanied by central bank accountability for the mandate given, for example, attainment of the inflation target under ITF. Central bank credibility is only achievable if the inflation target is attained. In this case, a consistent monetary policy response toward achievement of the inflation target is vital for credibility. Consistency is often measured in terms of certain policy rules, such as the Taylor rule, when setting the monetary policy rate. How much stronger central bank interest policy must be from the policy rules depends on the credibility. Weaker credibility, for example, will require a stronger monetary policy response. In other words, while the central bank builds credibility in terms of attaining the inflation target, the policy response must be tighter than stipulated

by the corresponding policy rule and, therefore, the impact on economic growth tends to be more pronounced.

The next level of central bank policy credibility is reputation (Barro & Gordon, 1983). Reputation is earned if the central bank can consistently build enough policy credibility with the public over a prolonged period. The central bank is said to have garnered a sound reputation if the public believes that the central bank has a proven track record in its avowed commitment to attaining the inflation target through an appropriate policy response. Consequently, public inflation expectations are anchored to the inflation target and, therefore, inflation bias due to policy inconsistency is avoided.

2.4.2. Transparency and Communication

In addition to central bank credibility and policy consistency, reputation is also required to strengthen transparency and communication with the public. Therefore, extending monetary policy transparency at the central bank is an inevitability. This phenomenon is congruent with the trend of greater central bank independence against the backdrop of a democratic government, application of ITF and the importance of anchoring expectations on the financial markets. In this case, several interesting observations can be found. *Firstly*, efforts have been poured into formulating a set of best practices to be used as a reference for central banks to strengthen monetary policy transparency (Blinder, 2001; IMF, 1999). The IMF, for example, assesses four critical aspects of policy transparency, namely: (1) clarity of roles, responsibilities and objectives of the monetary policy authority; (2) open process for formulating and reporting monetary policy decisions; (3) public availability of information on monetary policy; and (4) accountability and assurances of integrity from the monetary authority.

Secondly, a number of theories and analyses have been developed to provide formulae on transparency and its correlation with monetary policy, basic policy considerations and its impact on the economy. Fundamentally, the theories deal with asymmetric information in monetary policy, primarily in the context of the focus and clarity of the central bank's preferences in terms of the objectives and relevance of monetary policy as well as the influence of market expectations on the monetary policy response and transmission, including the debate between rules versus discretion for monetary management. *Thirdly*, although lacking in number, several studies have provided empirical evidence to show the positive contribution of increased central bank transparency on economic performance. Specifically, increased transparency can help to control inflation, reflected by the negative correlation between transparency and inflation. On the other hand, increased transparency feeds through lower output volatility, or at least does not exacerbate output volatility.

Greater transparency was demanded in the wake of the GFC in 2008/2009 and became a leading agenda item for central banks to implement monetary policy. An important development of central bank communication after the GFC was the emergence of forward guidance, defined as providing information and future monetary decisions for the market (Bihari, 2015). This practice was introduced by the US Federal Reserve at the announcement of the Federal Open Market Committee (FOMC) on December 16, 2008 as follows:

The Federal Reserve will employ all available tools to promote the resumption of sustainable economic growth and to preserve price stability. In particular, the Committee anticipates that weak economic conditions are likely to warrant exceptionally low levels of the federal funds rate for some time. The focus of the Committee's policy going forward will be to support the functioning of financial markets and stimulate the economy through open market operations and other measures that sustain the size of the Federal Reserve's balance sheet at a high level.

After the Fed, other central banks, including the BOJ, Bank of England and ECB, followed suit and announced forward guidance as a central bank practice.

Csortos, Lehmann, and Szalai (2014) analyzed the practice of forward guidance during a crisis and the experiences of the Fed and ECB. The study found that forward guidance is either forecast based or commitment based. In terms of the former, the central bank delivers forecast information based on currently available information and the most likely monetary policy response. Meanwhile, commitment-based forward guidance aims to influence through greater understanding of common temporary strategies, for example, the markets should be tolerant of higher inflation due to a low policy rate. In that context, the Fed and ECB apply commitment-based guidance, while the forward guidance of the BOJ and Bank of England tends to be more state-dependent.

Filardo and Hofmann (2014) reviewed the effectiveness of the forward guidance applied by the Fed, ECB, BOJ, and Bank of England. The study measured the impact of the policy rate on financial markets and economy from three aspects: (1) in terms of forward guidance implying a looser future monetary policy stance compared to market expectations, thus influencing short and long-term interest rates; (2) in terms of the central bank providing greater clarity on the future direction of the policy rate, thus the volatility of market expectations concerning future interest rates would be lower and could even lower the risk premium; and (3) in terms of forward guidance mentioning a specific indicator, which would leave the markets more sensitive to the data released for that indicator. Furthermore, although the effectiveness of forward guidance varies over time and between central banks, the study stressed that central bank credibility was the most important factor for the success of monetary policy instruments. To ensure effectiveness, the central bank must clearly communicate forward guidance as a trusted form of credible central bank commitment and interpreted by the markets as intended by the central bank.

Moving forward, transparency and communication will become increasingly important for the central bank and also academia. Evaluations and improvements must be undertaken by the central bank in terms of the monetary policy communication strategy and transparency, referring to international best practices, the monetary policy regime applied and current development of the financial markets. The issue of monetary policy transparency is also an interesting object for further study in academia. Moreover, a number of improvements are still required in terms of scientific and theoretical reviews linked to modeling and

analyzing the influence on economic performance, inflation and the central bank. Empirical studies are still required to analyze the impact of transparency on economic performance and financial market development.

2.4.3. Institutional Arrangements and Coordination

Central bank credibility requires strong institutional arrangements at the central bank and through coordination with the government and other relevant authorities. At the central bank, the policy framework must be strengthened in terms of formulating an appropriate policy response to attain the inflation target and financial stability based on macroeconomic projections and macrofinancial linkages, particularly between the banking and external sectors. The integrated analysis and analytical framework underlies formulation of the monetary, payment system and macroprudential policy mix. Research must be strengthened in terms of the effectiveness of the policy framework as well as the accuracy of macroeconomic and macrofinancial projections. Similarly, research and analysis are required to broaden understanding on macrofinancial linkages, procyclicality, and systemic risk. The data and statistics must also be strengthened, including the development of inflation, macroeconomic, payment system stability and financial system stability indicators, and even development of an early detection system, as well as cross-sector balance sheet statistics at the national and provincial levels.

The decision-making process must also be strengthened. A difference of opinion has emerged at various central banks, however, about whether it is more desirable to maintain or integrate the different committees for monetary policy and financial stability policy. Kohn (2015), for example, recommends maintaining the different committees, considering the disparate targets and focus, instruments, and accountability. This practice emerged at the Bank of England after the GFC in the form of establishing three committees outside of the monetary policy committee linked to financial stability, namely the Prudential Regulation Authority (PRA), created by the Bank of England for microprudential policy, the Financial Conduct Authority (FCA) for the financial markets, and the Financial Policy Committee for macroprudential policy. At BI, policy formulation and decisionmaking are conducted at the Board of the Governors meeting, concerning monetary, macroprudential and payment system policy. There are also three committees, namely monetary policy, financial stability, and the payment system, headed by the respective deputy government to oversee policymaking. A joint committee was also formed to strengthen the current policy mix, as well as the desired policy mix, through integrated analysis and projections from the three policy aspects. This differs considerably from the monetary policy framework based solely on ITF.

Central bank independence is never absolute. Independence is based on policy autonomy pursuant to prevailing laws. Therefore, the effectiveness of central bank policy depends on interaction and coordination with other policies, be it fiscal policy and the real sector by the government, microprudential regulation and supervision by a separate authority or conditions in the business community and financial sector. In the case of Indonesia, BI coordinates closely with the Government and other relevant authorities. Fiscal and monetary policy coordination is necessary

between BI and the Government when formulating the national budget and various other macroeconomic management aspects, nationally and locally. Although BI has been granted independence under law, the policy mix implemented represents an integral part of the national policy mix, including macroeconomic policy, financial system stability, and structural reforms (Warjiyo, 2013). In terms of financial system stability, pursuant to Law No. 9 of 2016 on Prevention and Resolution of Financial System Crises, national coordination is achieved through the Financial System Stability Committee (KSSK), headed by the Minister of Finance, with the Governor of BI, Chairman of the Financial Services Authority (OJK), and Chairman of the Deposit Insurance Corporation (LPS) as members.

2.5. Concluding Remarks

This chapter has discussed central bank evolution and reform since inception in the seventeenth century until after the GFC in 2008/2009 and several important lessons were discovered. *First*, the central bank plays a significant role in supporting sustainable economic growth by creating and maintaining price (and exchange rate) stability, while preserving financial system stability. This dual mandate has existed since the beginning of central banking but was reemphasized in the wake of the GFC. *Second*, to achieve the dual mandate, the central bank must be given jurisdiction for monetary policy (including management of foreign capital flows), the payment system as well as macroprudential regulation and supervision. Experience from the ITF era, however, revealed that a central bank focused solely on monetary policy allows financial instability to accumulate, thereby precipitating the GFC in 2008/2009. *Third*, the central bank needs autonomy to achieve its mandate. Experience from when central banks were under government control demonstrated the adverse effect on macroeconomic stability that led to economic crises. *Fourth*, institutional strengthening of the central bank is required to ensure that independence is accompanied by stronger public accountability and transparency.

How can a central bank build credibility and reputation when pursuing its responsibilities to achieve the dual mandate? Herein lies the importance of theoretical research and empirical studies on the roles of currency, credit, and foreign capital flows in the economy. Through a synthesis of Neoclassical and Keynesian theories, money supply influences inflation in the long term, although there is a short-term trade-off between inflation and economy growth. An efficient financial system is always in equilibrium; thus, currency and credit are perfect substitutes. Likewise, foreign capital flows move freely between countries and, therefore, fully flexible exchange rates are required. Consequently, central bank policy is represented by price stability through setting short-term interest rates similar to monetary policy based on ITF.

Nevertheless, the GFC of 2008/2009 reversed the prevailing thinking and assumptions. The financial system is not always in a state of equilibrium, financial frictions stem from information asymmetry, financial product innovation, evaluation methods, and risk management, as well as risk-taking behavior. Furthermore, the financial cycle tends to amplify the economic cycle by accelerating credit growth, asset price bubbles, and leverage, which heighten the risk of a crisis

occurring. Foreign capital flow mobility is also increasing, thus further amplifying the financial cycle and exchange rate volatility beyond fundamental values. Under such conditions, price stability alone is insufficient to maintain macroeconomic and financial system stability and, therefore, is unable to support sustainable economic growth. In addition, microprudential regulation and supervision to ensure the soundness of individual financial institutions would also not be able to guarantee financial system stability due to the factors that determine macrofinancial linkages in the amplification of the financial cycle.

Modern central bank policy since the GFC in 2008/2009 to achieve price (and exchange rate) stability as well as financial system stability is characterized by a mix of monetary policy (including foreign capital flow management), payment system policy, and macroprudential policy. In that context, theoretical understanding of the function of currency and credit in the economy, foreign capital flow mobility, and the exchange rate, as well as monetary policy transmission mechanisms, as discussed in Chapters 3–5, is crucial. The ITF-based policy framework discussed in Chapters 6–9 represents the foundation of the central bank's policy mix. Similarly, institutional reform through independence on one hand and stronger accountability, transparency and communication on the other is discussed in Chapters 10–12, which is constantly required to strengthen central bank credibility and reputation. The modern central bank policy mix framework since the GFC of 2008/2009 is discussed in full through Chapters 13–15 in terms of the theories and practices adopted at several central banks. It is important to note that policy mix theories and practices represent a new central banking paradigm that emerged after the GFC, and BI has pioneered such practices, especially since 2010.

This page intentionally left blank